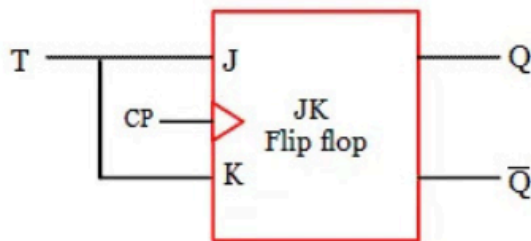


Circuit Simulation Lab:14

T Flip-flops

The purpose of this experiment is to introduce the design of simple sequential circuits, in this case T Flipflop. T flip flop is a modification of JK flip-flop. The J and K inputs are connected together to get the T input of flip flop. It is also called as Toggle flip flop

Logic Diagram



CP	T	Q_{n+1}	State
1	0	Q_n	NO CHANGE
1	1	$\bar{Q}_n$	TOGGLES

Behavior modeling

Code:

```
module tff(  
input t,  
input clk,  
input rst,  
output reg q,  
output qbar  
);
```

```
always @(posedge clk)
```

```
begin
```

```
if(rst)
```

```
    q <= 0;
```

```
else if(t)
```

```
    q <= ~q;
```

```
else
```

```
    q <= q;
```

```
end
```

```
assign qbar = ~q;
```

```
Endmodule
```

```
Testbench
```

```
module tb_tff;
```

```
reg t,clk,rst;
```

```
wire q,qbar;
```

```
tff dut(
```

```
.t(t),
```

```
.clk(clk),
```

```
.rst(rst),
```

```
.q(q),
```

```
.qbar(qbar)
```

```
);
```

```
always #10 clk = ~clk;
```

```
initial begin
```

```
$dumpfile("tff.vcd");
```

```
$dumpvars(0,tb_tff);
```

```
clk = 0;
```

```
rst = 1;
```

```
t = 0;
```

```
#20;
```

```
rst = 0;
```

```
// No Change
```

```
t = 0; #20;
```

```
// Toggle
```

```
t = 1; #20;
```

```
// Toggle Again
```

```
t = 1; #20;
```

```
// Hold
```

```
t = 0; #20;
```

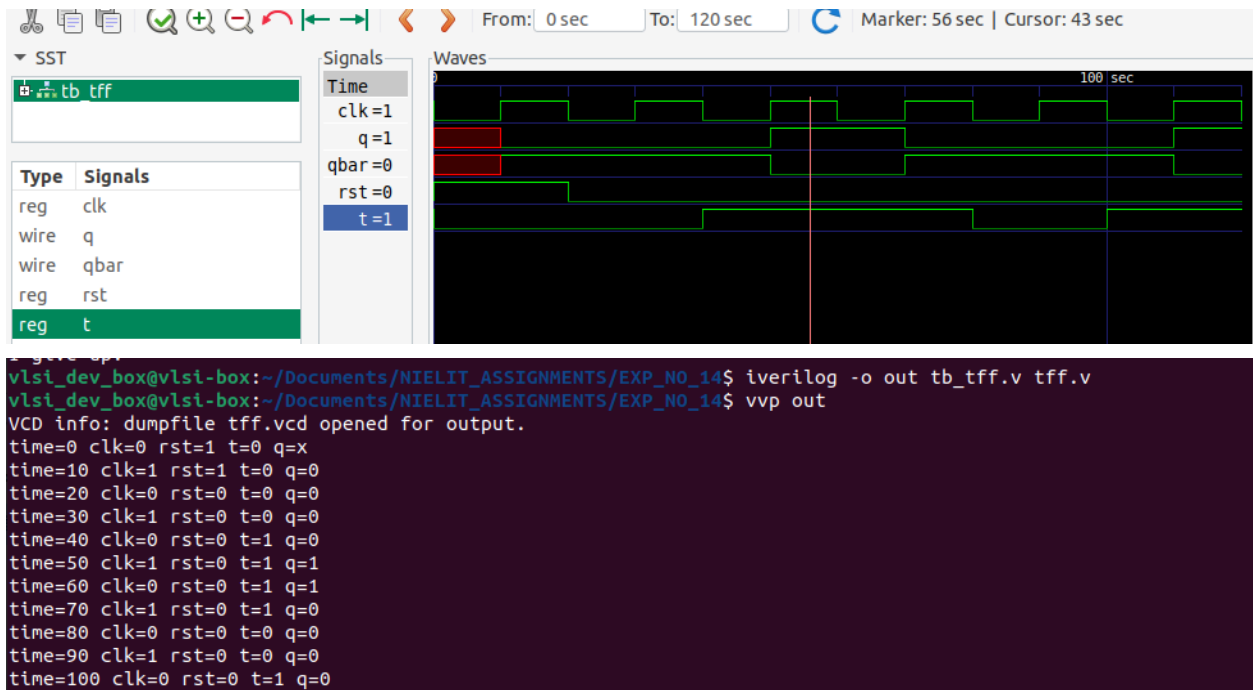
```
// Toggle
t = 1; #20;

$finish;

end

initial
$monitor("time=%0t clk=%b rst=%b t=%b q=%b",
        $time,clk,rst,t,q);

Endmodule
```



Structural modeling (Use the JK Flip Flop for designing the T FF)

```
module tff_using_jk(
input t,
input clk,
output q
);

jk jk1(
.clk(clk),
```

```
.j(t),  
.k(t),  
.q(q)  
);
```

```
Endmodule
```

```
Tb:
```

```
module tb_tff_using_jk;
```

```
reg t,clk;
```

```
wire q;
```

```
tff_using_jk dut(
```

```
.t(t),
```

```
.clk(clk),
```

```
.q(q)
```

```
);
```

```
always #10 clk = ~clk;
```

```
initial begin
```

```
$dumpfile("tff_jk.vcd");
```

```
$dumpvars(0,tb_tff_using_jk);
```

```
clk = 0;
```

```
// Hold
```

```
t = 0; #20;
```

```
// Toggle
```

```
t = 1; #20;
```

```
// Toggle
```

```
t = 1; #20;
```

```
// Hold
```

```
t = 0; #20;
```

```
// Toggle
```

```
t = 1; #20;
```

```
// Hold
```

```

t = 0; #20;

$finish;

end

initial
$monitor("time=%0t clk=%b t=%b q=%b",
        $time,clk,t,q);

Endmodule

```

```

vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_14$ iverilog -o out tff_using_jk.v tb_tff_using_jk.v jk.v
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_14$ vvp out
VCD info: dumpfile tff_jk.vcd opened for output.
time=0 clk=0 t=0 q=x
time=10 clk=1 t=0 q=x
time=20 clk=0 t=1 q=x
time=30 clk=1 t=1 q=x
time=40 clk=0 t=1 q=x
time=50 clk=1 t=1 q=x
time=60 clk=0 t=0 q=x
time=70 clk=1 t=0 q=x
time=80 clk=0 t=1 q=x
time=90 clk=1 t=1 q=x
time=100 clk=0 t=0 q=x
time=110 clk=1 t=0 q=x
time=120 clk=0 t=0 q=x

```